

Case Study – RAG + Generative AI Service

Role: Mid-Level Developer

Unified Business Scenario & Global Constraints

Build a multi-turn RAG QA + generative service over the internal knowledge base (employee handbook, compliance guides, technical specifications, architecture documents). The corpus is bilingual (CN/EN) and includes a small portion of scanned PDFs. Candidates may choose any tech stack, but all key technical choices must be justified in the deliverables with clear, quantitative evidence and validation.

Global Constraints (apply to this role unless overridden below)

1. Performance
 - 90% of QA requests must complete end-to-end within 10 seconds; support ≥ 5 concurrent requests on a single instance.
2. Cost
 - Provide token-cost estimates per 1,000 calls, and explain your model-version selection rationale with explicit trade-offs (quality, cost, latency).
3. Quality Metrics (must be quantified)
 - RAG: Faithfulness ≥ 0.85 (define your evaluation method); Context Precision ≥ 0.70 .
 - Generative answers: define the evaluation standard and implementation. Target thresholds: Answer Compliance $\geq 80\%$; Style Consistency $\geq 80\%$; Refusal Appropriateness $\geq 80\%$.
4. Logging & Tracing
 - Emit structured logs sufficient for generation monitoring and issue diagnosis.
5. Security
 - Minimal prompt-injection defenses; basic PII handling; all answers must strictly ground to retrieved context.

Objective

Extend and harden the service to include configurable retrieval modes, reranking, generative quality controls, complete observability, caching, and reproducible diagnosis.

Functional Requirements

1. 1) Retrieval Controls
 - Support at least two retrieval modes: vector-only and hybrid. Allow enabling/disabling reranker via configuration (no code change).
2. 2) Refusal & Safety Handling
 - When confidence is low, the query is out-of-scope, or safety rules trigger, return a refusal with guidance.
3. 3) Privacy
 - Apply basic PII redaction to outputs and logs.
4. 4) Minimal Operations Report
 - Produce a simple text/CSV report including: p50/p95 latency, token usage, cache hit rate, refusal rate, and answer compliance rate.

Non-Functional & Quantitative Metrics

1. 1) Retrieval Quality
 - Compare three configurations: vector-only, hybrid, hybrid+rerank. Provide quantitative results and conclusions
2. 2) Evolvability
 - Design for iterative optimization: retrieval strategy changes, model version swaps, metric/logging enhancements.
3. 3) Generative Quality (advanced targets)
 - Answer Compliance $\geq 90\%$; Refusal Appropriateness $\geq 90\%$; Style Consistency ≥ 0.85 (describe your evaluation method).
4. 4) Issue Diagnosis
 - Document at least two issues (e.g., compliance drop, refusal spike) with log/metric evidence, fix rationale, and post-fix improvement $\geq 10\%$.

Deliverables

- Complete code and configs
- One-click evaluation script.
- Evaluation report with before/after comparisons.
- Log field dictionary and sample logs.